DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

Eigen Values and CHT

Lecture No. 01



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Recap of previous lecture



Topic HOMOGENEOUS S.O.Eq.

Topics to be Covered



Topic EIGEN VALUES & EIGEN VECTORS

Q. Consider $A_{m\times n} X_{n\times 1} = B_{m\times 1}$ then which is false ?

(a) if $m > n$ & $B \neq 0$ & $\rho(A) < \rho(A:B)$ then system has No solution (T)

(b) if $m = 5$, $n = 3$, $B = 0$, & $\rho(A) = 3$ then system has only Zero solution. (T)

(c) if $m = 3$, $n = 5$, $B = 0$, then system has Non Trivial solutions. (T)

(d) if $m = n$, $B = 0$, $|A| = 0$ then system has Non Zero solutions. (T)

(e) if $m > n$, $B = 0$ then system has solution (T) → consistent.

(f) if $m > n$, $B = 0$ then system has Multiple solutions (f)

(g) if $m > n$, $B = 0$, $\rho(A) < n$ then system has Multiple solutions. (T)
($\approx \infty$ sol)

EIGEN VALUES & EIGEN VECTORS

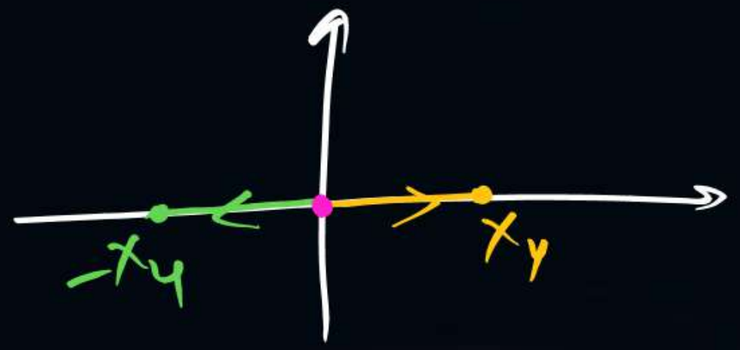
Consider $A = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}$ & let $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \hat{i} + \hat{j}$, $\|X_1\| = \sqrt{1+1} = \sqrt{2}$

Now $AX_1 = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 3 \end{bmatrix} = X_2 = -3\hat{i} + 3\hat{j}$, $\|X_2\| = \sqrt{9+9} = 3\sqrt{2}$



Now Taking $X_3 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, then $AX_3 = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix} = 3\begin{bmatrix} -1 \\ 2 \end{bmatrix} = 3X_3$

Again taking $X_4 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ then $AX_4 = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -\begin{bmatrix} 1 \\ 0 \end{bmatrix} = -X_4$



$AX_5 = \cdots\cdots = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 = 0 \cdot X_5$

Note Let us take $X_6 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ then

$AX_6 = \begin{bmatrix} -1 & -2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \lambda \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \lambda X_6$

it is obviously true & Nothing special in this so X_6 can not be taken as E. Vector i.e E. Vector is always Non Zero Vector

Def^n: Consider the Square Mat $A_{n\times n}$ then Non Zero vector X is called Eigen Vector corresponding to E.Value λ (Real / Complex) if we are able to find a relationship of the type

$$AX = \lambda X$$

Here λ = special Value
X = special Vector

(LHS is the Multi of Mat & Vector) = (RHS is the Scalar Multi in Vector)
(Tough job) (Easy job)

eg find the EValues of $A = \begin{bmatrix} 2 & -3 & 1 \\ 4 & 1 & 2 \end{bmatrix}_{2\times 3}$ (senseless Question)

Method of Calculating E-Values :→

Let given Mat is $A_{n\times n}$ & λ is the E-Value & X is it's E-vector then

$$AX = \lambda X \quad \text{---} (1)$$

$$AX - \lambda X = 0$$

$$(A - \lambda I)X = 0$$

$$MX = 0 \quad \text{---} (2)$$

↓

Non Zero E-Vector / Non zero Solution / ∞ Soln.

& the condition for the existence of ∞ Soln will be

$$\rho(M) < n \text{ or } |M| = 0$$

$$\Rightarrow \rho(A - \lambda I) < n \text{ or } |A - \lambda I| = 0 \quad \text{---} (3)$$

equⁿ (3) is called Characteristic equⁿ of A & Roots of this equⁿ (i.e values of λ) are called E-Values / E-Roots / Char Values / Char Roots / Latent Roots / special Values.

Q: Find the E. Values of ① $A=\begin{bmatrix}4&2\\2&4\end{bmatrix}$ ② $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$, ③ $A=\begin{bmatrix}0&-1\\1&0\end{bmatrix}$

Sol:

$$A-\lambda I=\begin{bmatrix}4&2\\2&4\end{bmatrix}-\lambda\begin{bmatrix}1&0\\0&1\end{bmatrix}$$

$$=\begin{bmatrix}(4-\lambda)&2\\2&(4-\lambda)\end{bmatrix}$$

= Char. Matrix

Now C. Equⁿ is $|A-\lambda I|=0$

$$\begin{vmatrix}(4-\lambda)&2\\2&(4-\lambda)\end{vmatrix}=0$$

$\lambda^2-8\lambda+12=0 \Rightarrow \lambda=6,2$

C. Equⁿ

③ $A=\begin{bmatrix}0&-1\\1&0\end{bmatrix}$, Soo

$$A-\lambda I=\begin{bmatrix}0&-1\\1&0\end{bmatrix}-\lambda\begin{bmatrix}1&0\\0&1\end{bmatrix}=\begin{bmatrix}-\lambda&-1\\1&-\lambda\end{bmatrix}$$

C. Equⁿ is $|A-\lambda I|=0$

$$\begin{vmatrix}-\lambda&-1\\1&-\lambda\end{vmatrix}=0 \Rightarrow \lambda^2+1=0$$

$\lambda^2=-1$

$\lambda=\pm\sqrt{-1}=\pm i$

i.e. E. Values are i & $-i$

③ $A=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}$ then $A-\lambda I=\begin{bmatrix}2&1&1\\1&2&1\\0&0&1\end{bmatrix}-\lambda\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}(2-\lambda)&1&1\\1&(2-\lambda)&1\\0&0&(1-\lambda)\end{bmatrix}$

C.Equ^n is $|A-\lambda I|=0$

$$\begin{vmatrix}(2-\lambda)&1&1\\1&(2-\lambda)&1\\0&0&(1-\lambda)\end{vmatrix}=0$$

- - - - - - -

$$(\lambda-1)^2(\lambda-3)=0$$

$\lambda=1,1,3$

Note:

Q. Find E. Values of $A=\begin{bmatrix}&&&&\\&&&&\\&&&&\\&&&&\\&&&&\end{bmatrix}_{5\times5}$

In order to solve such types of Questions we should have an idea about Properties of E. Values ?

Trace = $\sum$ = sum of E Values

$|A|=3$ = Product of E Values.

PROPERTIES of E. Values → Let $A_{n\times n}$ is the given Matrix

& let $\lambda_1, \lambda_2, \lambda_3, \ldots\ldots, \lambda_n$ are it's E. Values.

① Number of E. Values of A = order of A (whether distinct or Repeated)

② Sum of E. Values = Trace (A) i.e $\lambda_1+\lambda_2+\lambda_3+\ldots\ldots+\lambda_n = Tr(A)$

③ Product of E Values = Det (A) i.e $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 \ldots\ldots \cdot \lambda_n = |A|$

④ (Zero is an E. Value of A) iff (Mat is Singular) i.e $\lambda = 0 \iff |A| = 0$

⑤ E Values of U.T.M, L.T.M, Diagonal Mat, Scalar Mat are just the Diag elements

eg $A = \begin{bmatrix} 2 & -1 & 0 & 4 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 4 \end{bmatrix} \Rightarrow \lambda = 2, 3, 0, 4$

② $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \Rightarrow \lambda = 1, -1, 2, 4$

(6) W.K.T that EValues of $A_{n\times n}$ are $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ then

(i) EValues of A^T are same i.e $\lambda_1, \lambda_2, \lambda_3 \ldots, \lambda_n$

(ii) EValues of A^m are $\lambda_1^m, \lambda_2^m, \lambda_3^m, \ldots, \lambda_n^m$ (where $m \in N$)

(iii) EValues of A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{1}{\lambda_3}, \ldots, \frac{1}{\lambda_n}$ (Provided $|A| \neq 0$)

(iv) EValues of (adj A) are $\frac{|A|}{\lambda_1}, \frac{|A|}{\lambda_2}, \frac{|A|}{\lambda_3}, \ldots, \frac{|A|}{\lambda_n}$

Proof (ii)

$AX = \lambda X$ → EValue of A

$A(AX) = \lambda(AX)$

$A^2X = \lambda(\lambda X)$

$A^2X = \lambda^2 X$ → EValue of A^2

Proof (iii)

$AX = \lambda X$ → EValue of A

$\lambda X = AX$

$\lambda(A^{-1}X) = A^{-1}(AX)$

$A^{-1}X = \left(\frac{1}{\lambda}\right)X$ → EValue of A^{-1}

Proof (iv)

$AX = \lambda X$ → E value of A

$A^{-1}(AX) = \lambda(A^{-1}X)$

$IX = \lambda\left[\frac{(\text{adj } A)X}{|A|}\right]$

$\left(\frac{1}{\lambda}\right)X = \left(\frac{\text{adj } A}{|A|}\right)X$

$\left(\frac{|A|}{\lambda}\right)X = (\text{adj } A)X$

or $(\text{adj } A)X = \left(\frac{|A|}{\lambda}\right)X$ → E value of (adj A)

Q: GATE If Evalues of $A=\begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$ are 4 & 8 then $x+y=?$

Sol: w.k that $\lambda_1+\lambda_2 = Tr(A)$

$4+8 = 2+y$

$\Rightarrow \boxed{y=10}$

Again $\lambda_1 \cdot \lambda_2 = |A|$

$(4)(8) = 2y-3x$

$32 = 20-3x \Rightarrow \boxed{x=-4}$

$\therefore$ Ans $x+y=6$ Ans

Q: GATE If Trace & Det of $A_{2\times 2}$ are -2 & -35 resp then find $\lambda_1^2+\lambda_2^2=?$

Sol: As Q, M-I

$\lambda_1+\lambda_2 = Tr(A)$ & $\lambda_1 \cdot \lambda_2 = |A|$

$\lambda_1+\lambda_2 = -2$ ——(1) & $\lambda_1 \cdot \lambda_2 = -35$ ——(2)

$\Rightarrow \lambda_1 = 5$ & $\lambda_2 = -7$

$\therefore \lambda_1^2+\lambda_2^2 = (5)^2+(-7)^2$

$= 25+49 = 74$ Ans

M-II $\lambda_1+\lambda_2=-2$, $\lambda_1\lambda_2=-35$ — (1)

we know that

$$(\lambda_1-\lambda_2)^2=(\lambda_1+\lambda_2)^2-4\lambda_1\lambda_2$$

$$=(-2)^2-4(-35)$$

$$=4+140=144$$

$\lambda_1-\lambda_2=12$ — (2)

$$2\lambda_1=10\Rightarrow\lambda_1=5$$

$$\lambda_2=5-12=-7$$

} Ans

2025 GATE (CE)

if $A=\begin{bmatrix}1&2&-3\\4&0&5\\-3&2&1\end{bmatrix}$ then $A^T=?$

M-III $(\lambda_1+\lambda_2)^2=\lambda_1^2+\lambda_2^2+2\lambda_1\lambda_2$

$$(-2)^2=\lambda_1^2+\lambda_2^2+2(-35)$$

$$\lambda_1^2+\lambda_2^2=4+70=74$$

Q: if P & Q are two square Matrices of same order s.t $PQ = I$ $\Rightarrow$ $P^{-1} = Q$, $Q^{-1} = P$

then 0 is an E-Value of

(a) P But Not Q

(b) Q " " P

(c) Both P & Q

(d) ✓ Neither P Nor Q

i.e Both P & Q are Invertible

$\Rightarrow$ " " " are Non Singular

$\Rightarrow$ 0 Can not be the E Value of any Mat.

Q: If $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & \frac{-1+i\sqrt{3}}{2} & 0 \\ 4 & 8 & \frac{-1-i\sqrt{3}}{2} \end{bmatrix}$ then $Tr(A^{102}) = ?$

Sol: $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & \omega & 0 \\ 4 & 8 & \omega^2 \end{bmatrix}$ = L.T.M

$\Rightarrow \lambda = 1, \omega, \omega^2$

Note ~~$Tr(A) = 1+\omega+\omega^2 = 0$~~

~~$\therefore Tr(A^{102}) = (Tr(A))^{102}$~~

~~Blunder $= 0^{102} = 0$~~

E values of A are $1, \omega, \omega^2$

" " A^{102} are $1^{102}, \omega^{102}, (\omega^2)^{102}$

$= 1, (\omega^3)^{34}, \omega^{204}$

$= 1, (\omega^3)^{34}, (\omega^3)^{68}$

E values of $A^{102} = 1, 1^{34}, 1^{68}$

$= 1, 1, 1$

$\therefore Tr(A^{102}) = 1+1+1 = 3$

Q. $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ then

→ EValues of $A = 2, -1, 0, 3$

" " $A^T = 2, -1, 0, 3$

" " $A^3 = 8, -1, 0, 27$

" " $A^{-1} = \frac{1}{2}, -1, \text{DNE}, \frac{1}{3}$

(Blunder)^2

" " $A^{-1} = ?$ = senseless Question.

$\therefore A^{-1} = \text{DNE}.$

THANK - YOU

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